



Your Ultimate AI Study Companion

Transforming the way students prepare for exams using the power of Generative AI.

The Problem

Students are overwhelmed by unstructured information.

-  **Time-Consuming:** Hours spent manually summarizing long lectures and textbooks.
-  **Inefficient:** Passive reading instead of active recall techniques.
-  **Overwhelming:** Lack of structured study plans leads to cramming and anxiety.
-  **Scattered Tools:** Managing multiple separate apps for notes, flashcards, and diagrams.

The Solution: ExamNotes AI

A centralized, premium AI-driven learning platform.

By leveraging **Google's Gemini AI**, we instantly convert any topic into actionable, structured study materials.

-  **Instant Generation:** From topic query to full study material in seconds.
-  **Active Recall:** Built-in smart flashcards to drastically improve memory retention.
-  **Visual Learning:** Auto-generated charts and mind maps for complex concepts.

✨ Key Features

- 📝 **AI-Powered Notes:** Detailed, highly-structured notes generated instantly.
- 🗺️ **Study Roadmaps:** Step-by-step guides to master any subject from scratch.
- 📁 **Smart Flashcards:** 3D flippable flashcards tailored specifically to the generated topic.
- 📊 **Visual Charts:** Integrated Mermaid.js diagrams and flowcharts for visual learners.
- 📄 **PDF Export:** Professionally formatted downloads for offline, distraction-free study.

How It Works (Under the Hood)

A Seamless & Fast User Journey

1. **Input:** User enters a topic or subject they need to learn.
2. **Process:** Our secure Express backend queries the Google Gemini API to intelligently structure the educational content.
3. **Visualize:** The React (Vite) frontend renders the payload into beautiful notes, interactive flashcards, and Mermaid diagrams.
4. **Export:** User downloads the materials via PDFKit or saves them to their personalized dashboard.

Architecture & Tech Stack

Built for Speed, Scale, and a Premium Experience

- **Frontend:** React 19, Vite, Tailwind CSS 4.0, Motion (Glassmorphic UI design)
- **Backend:** Node.js, Express.js
- **Database:** MongoDB Atlas (Mongoose)
- **AI Engine:** Google Gemini SDK
- **Infrastructure:** JWT Auth + Firebase Admin, PDFKit, Mermaid.js, Recharts

Business Model

Freemium Model powered by a Custom Credit System

- **Free Tier:** Basic credits granted upon signup to experience the core AI magic.
- **Premium Tier:** Seamless **Stripe** integration allows users to purchase top-up credit packs.
- **Premium Features:** Access to unlimited generations, high-quality PDF exports, and advanced visual flowcharts.

Future Roadmap

- **Phase 1 (MVP - Current):** AI notes generation, interactive flashcards, and basic diagrams.
- **Phase 2 (File Processing):** PDF/Document upload capabilities (Chat with your textbook).
- **Phase 3 (Social):** Collaborative study rooms and peer-to-peer flashcard sharing.
- **Phase 4 (Expansion):** Native Mobile App release for on-the-go learning.

Thank You

ExamNotes AI

Study Smarter, Not Harder.

Links:

-  [Live Prototype Demo](#)
-  [GitHub Repository](#)